STARTUP PREDICTION

Predicting whether a Startup will be acquired or shut down

- **Team Number:** 14

- **Team members:** Prameya Krishn - Documentation Lead
  Veronica Silvery - PPT Lead
  Meesala Jyothi Prakash - Video Lead
  Rohit Raj - Executive Lead

- **GitHub Repository: https://github.com/2026dab1301-pKrishn/AI_101_Final_Project_Group-14**

PROBLEM STATEMENT & DATASET

THE PROBLEM:

Startups face uncertainty about their future outcomes. Some startups are acquired by other companies, while others shut down.

OUR OBJECTIVE:

To develop a Logistic Regression model that predicts whether a startup will be acquired or closed based on its total funding amount.

DATASET INFORMATION:

- Dataset: Startup Investments Analysis
- Source: Hugging Face
  Original: Crunchbase (via Kaggle)
- Records: 4,757 | Columns: 23
- Target Variable: Status
- 1 = Acquired | 0 = Closed
- Missing Values: 0
- Duplicate Records: 18

PROJECT APPROACH

01
DATASET

Startup data

02
FEATURE

Select funding and status

03
PREPROCESS

Log transform and Scaling

04
MODEL

Logistic Regression

05
EVALUATION

Test and Predict Outcomes

DATASET → FEATURE SELECTION → PREPROCESSING → MODEL TRAINING → EVALUATION

TECHNOLOGIES USED:

Python | Pandas | NumPy | Matplotlib | Seaborn | Scikit-learn

RESULTS & ANALYSIS

Startup Status Distribution

Visualisation_analysis.ipynb

Visualisation_analysis.ipynb > status_counts = df["status"].value_counts().sort_index()

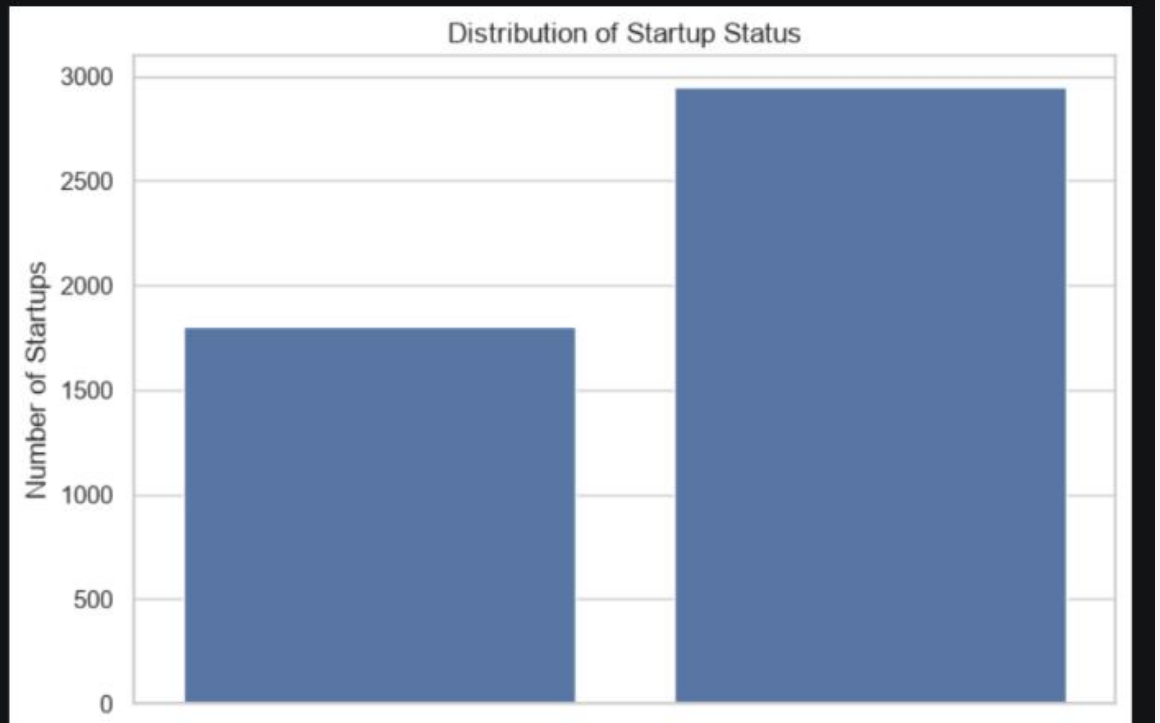
Generate | Code | Markdown | Colab | Run All | Clear All Outputs | Outline

```
status_counts = df["status"].value_counts().sort_index()

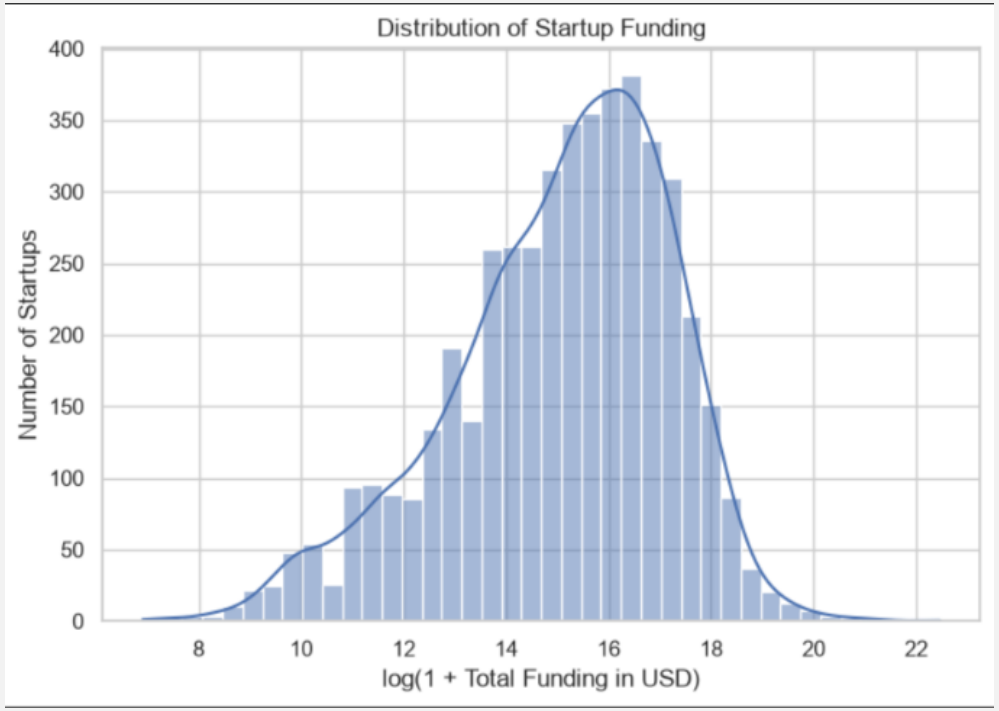
plt.figure(figsize=(7, 5))
sns.barplot(x=status_counts.index.astype(str), y=status_counts.values)

plt.title("Distribution of Startup Status")
plt.xlabel("Startup Status Code")
plt.ylabel("Number of Startups")
plt.tight_layout()
plt.show()
```

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Distribution of Total Startup Funding



MODEL PERFORMANCE:

- Testing Accuracy: 73.95%
- Baseline Accuracy: 61.97%
- 5-Fold CV Mean Accuracy: 72.27%

LEARNINGS & CONCLUSION

WHAT WE LEARNED:

- Data visualisation helps us understand patterns in startup data.
- Funding distribution is uneven across startups.
- Logistic Regression gave us a baseline for predicting startup outcomes

WHAT WORKED:

- Pandas helped us inspect and organise the dataset.
- Visualisations helped us explore startup status and funding patterns.
- Log transformation and Standard Scaling helped prepare funding data for model training.

WHAT DIDN'T:

- The Logistic Regression model struggled to identify startups in status class 0.

AI USE DISCLOSURE : AI tools were used for guidance, code assistance, and presentation preparation. The team reviewed and adapted the outputs for the project.

INDIVIDUAL CONTRIBUTIONS:

Rohit Raj : Dataset preparation and preprocessing
Prameya krishn : Model training
Meesala Jyothi Prakash : Model evaluation
Veronica Silvery : Visualisation & Analysis